Lab 1: DevOps Foundations & Continuous Integration

Name: Nitesh Sharma | GitHub: github.com/niteshcore | Repo: github.com/niteshcore/devops-lab1

2. Set up version control using Git

5 marks

Task: Initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

Steps / Commands performed:

- Initialize repository:

```
$ git init -b main
```

- Stage and commit changes:

```
$ git add README.md logs/
$ git commit -m "Initial commit"
$ git add app.py
$ git commit -m "Add app.py"
```

- Create a branch and commit on it:

```
$ git branch feature-login
$ git checkout feature-login
$ git add login.py && git commit -m "Add login feature"
```

- Merge branch into main:

```
$ git checkout main
$ git merge feature-login --no-ff
$ git log --oneline --graph --all
```

- Push to GitHub:

```
$ git remote add origin https://github.com/niteshcore/devops-lab1.git
$ git push -u origin main feature-login
```

Screenshot: init, stage, commit, branch, merge

```
Terminal — Dev-ops Lab

nitesh@macbook Dev-ops Lab % git init -b main
Initialized empty Git repository in /Users/niteshsharma/Dev-ops Lab/.git/
nitesh@macbook Dev-ops Lab % git status --short
?? README.md
?? logs/
nitesh@macbook Dev-ops Lab % git add README.md logs/task1-git-install-ubuntu.txt
nitesh@macbook Dev-ops Lab % git commit -m "Initial commit: README and Task 1 Git install log"
[main (root-commit) 9fc605a] Initial commit: README and Task 1 Git install log
 2 files changed, 47 insertions(+)
 create mode 100644 README.md
 create mode 100644 logs/task1-git-install-ubuntu.txt
nitesh@macbook Dev-ops Lab % git add app.py
nitesh@macbook Dev-ops Lab % git commit -m "Add app.py"
[main d9c22cb] Add app.py
 1 file changed, 1 insertion(+)
 create mode 100644 app.py
nitesh@macbook Dev-ops Lab % git branch feature-login
nitesh@macbook Dev-ops Lab % git checkout feature-login
Switched to branch 'feature-login'
nitesh@macbook Dev-ops Lab % git add login.py
nitesh@macbook Dev-ops Lab % git commit -m "Add login feature"
[feature-login 8ac0b59] Add login feature
 1 file changed, 2 insertions(+)
 create mode 100644 login.py
nitesh@macbook Dev-ops Lab % git branch
* feature-login
  main
nitesh@macbook Dev-ops Lab % git checkout main
Switched to branch 'main'
nitesh@macbook Dev-ops Lab % git merge feature-login --no-ff -m "Merge branch feature-login into main"
Merge made by the 'ort' strategy.
 login.py | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 login.py
nitesh@macbook Dev-ops Lab % git log --oneline --graph --all
*   517f4be Merge branch feature-login into main
|\
| * 8ac0b59 Add login feature
|/
* d9c22cb Add app.py
* 9fc605a Initial commit: README and Task 1 Git install log
```

Screenshot: push to GitHub

```
Terminal — Dev-ops Lab

nitesh@macbook Dev-ops Lab % git remote add origin https://github.com/niteshcore/devops-lab1.git
nitesh@macbook Dev-ops Lab % git remote -v
origin    https://github.com/niteshcore/devops-lab1.git (fetch)
origin    https://github.com/niteshcore/devops-lab1.git (push)
nitesh@macbook Dev-ops Lab % git push -u origin main feature-login
To https://github.com/niteshcore/devops-lab1.git
 * [new branch]      main -> main
 * [new branch]      feature-login -> feature-login
branch 'main' set up to track 'origin/main'.
branch 'feature-login' set up to track 'origin/feature-login'.
```

GitHub Repository: https://github.com/niteshcore/devops-lab1
Result: Repository created, commits made, feature-login branch created and merged into main, and both branches pushed to GitHub.